

1-1 LED BLINK

1sec ke liye ek on or ek off .

```
import RPi.GPIO as GPIO
import time

GPIO.setmode(GPIO.BOARD)

led = 3

GPIO.setup(led, GPIO.OUT)

while True:
    GPIO.output(led, True)
    time.sleep(1)

    GPIO.output(led, False)
    time.sleep(1)
```

2-Blink the two LEDs alternately with a delay of 0.5 seconds.

```
import RPi.GPIO as GPIO
import time

GPIO.setmode(GPIO.BOARD)

led1 = 3
led2 = 7

GPIO.setup(led1, GPIO.OUT)
GPIO.setup(led2, GPIO.OUT)

while True:
```

```
GPIO.output(led1, True)
time.sleep(0.5)
```

```
GPIO.output(led1, False)
time.sleep(0.5)
```

```
GPIO.output(led2, True)
time.sleep(0.5)
```

```
GPIO.output(led2, False)
time.sleep(0.5)
```

3-Question:

2 LEDs ko 1 second ke liye ek saath ON aur phir OFF karne ka Python program likho.

```
import RPi.GPIO as GPIO
import time
```

```
GPIO.setmode(GPIO.BOARD)
```

```
led1 = 3
```

```
led2 = 7
```

```
GPIO.setup(led1, GPIO.OUT)
```

```
GPIO.setup(led2, GPIO.OUT)
```

```
while True:
```

```
    GPIO.output(led1, True)
    time.sleep(0.5)
```

```
    GPIO.output(led2, True)
    time.sleep(0.5)
```

```
GPIO.output(led1, False)
time.sleep(0.5)
```

```
GPIO.output(led2, False)
time.sleep(0.5)
```

4-Practical Question

4 LEDs ko 0.5 second ke interval par ek-ek karke ON aur OFF karne ka Python program likho.

```
import RPi.GPIO as GPIO
import time
```

```
GPIO.setmode(GPIO.BOARD)
```

```
led1 = 3
led2 = 7
led3 = 11
led4 = 15
```

```
GPIO.setup(led1, GPIO.OUT)
GPIO.setup(led2, GPIO.OUT)
GPIO.setup(led3, GPIO.OUT)
GPIO.setup(led4, GPIO.OUT)
```

```
while True:
```

```
    GPIO.output(led1, True)
    time.sleep(0.5)
    GPIO.output(led1, False)
    time.sleep(0.5)
```

```
    GPIO.output(led2, True)
    time.sleep(0.5)
```

```
GPIO.output(led2, False)
time.sleep(0.5)
```

```
GPIO.output(led3, True)
time.sleep(0.5)
GPIO.output(led3, False)
time.sleep(0.5)
```

```
GPIO.output(led4, True)
time.sleep(0.5)
GPIO.output(led4, False)
time.sleep(0.5)
```

5-Practical Question

4 LEDs ko 2-2 ke pair mein 0.5 second ke interval par ON/OFF karne ka Python program likho.

```
import RPi.GPIO as GPIO
import time
```

```
GPIO.setmode(GPIO.BOARD)
```

```
led1 = 3
led2 = 7
led3 = 11
led4 = 15
```

```
GPIO.setup(led1, GPIO.OUT)
GPIO.setup(led2, GPIO.OUT)
GPIO.setup(led3, GPIO.OUT)
```

```
GPIO.setup(led4, GPIO.OUT)
```

```
while True:
```

```
    GPIO.output(led1, True)
```

```
    GPIO.output(led3, True)
```

```
    GPIO.output(led2, False)
```

```
    GPIO.output(led4, False)
```

```
    time.sleep(0.5)
```

```
    GPIO.output(led1, False)
```

```
    GPIO.output(led3, False)
```

```
    GPIO.output(led2, True)
```

```
    GPIO.output(led4, True)
```

```
    time.sleep(0.5)
```

6-Practical Question

4 LEDs ko alternate pair mein 1 second ke interval par ON/OFF karne ka Python program likho.

```
import RPi.GPIO as GPIO
```

```
import time
```

```
GPIO.setmode(GPIO.BOARD)
```

```
led1 = 3
```

```
led2 = 7
```

```
led3 = 11
```

```
led4 = 15
```

```
GPIO.setup(led1, GPIO.OUT)
```

```
GPIO.setup(led2, GPIO.OUT)
```

```
GPIO.setup(led3, GPIO.OUT)
```

```
GPIO.setup(led4, GPIO.OUT)
```

```
while True:
```

```
    GPIO.output(led2, True)
```

```
    GPIO.output(led4, True)
```

```
    GPIO.output(led1, False)
```

```
    GPIO.output(led3, False)
```

```
    time.sleep(1)
```

```
    GPIO.output(led2, False)
```

```
    GPIO.output(led4, False)
```

```
    GPIO.output(led1, True)
```

```
    GPIO.output(led3, True)
```

```
    time.sleep(1)
```

7-Seven segment

CLK-16

DID-18

VCC-4

GND-14

```
import sys
import os
import datetime
import time
import RPi.GPIO as GPIO
import tm1637

Display = tm1637.TM1637(23, 24, tm1637.BRIGHT_TYPICAL)

Display.Clear()
Display.SetBrightness(1)

while True:
    now = datetime.datetime.now()

    hour = now.hour
    minute = now.minute
    second = now.second

    current_time = [
        int(hour / 10),
        hour % 10,
        int(minute / 10),
        minute % 10
    ]

    Display.Show(current_time)
    Display.ShowDoublePoint(second % 2)

    time.sleep(1)
```

8-Print your name

GND-6

VCC-2

SDA-3

SCL-5

```
from RPLCD.i2c import CharLCD
```

```
from time import sleep
```

```
lcd = CharLCD(
```

```
    'PCF8574',
```

```
    0x27,
```

```
    cols=16,
```

```
    rows=2
```

```
)
```

```
lcd.clear()
```

```
lcd.write_string("Hello Aman!")
```

```
lcd.cursor_pos = (1, 0)
```

```
lcd.write_string("LCD working")
```

```
sleep(10)
```

```
lcd.clear()
```

Open terminal

Enter command

```
sudo raspi-config
```

3.interface option

I5 I2C SELECT

YES

OK

9 ABOUT RASPI-CONFIG

```
sudo apt-get install -y i2c-tools
```

```
sudo i2cdetect -y 1
```